CSA: Compressed Sparse Attention

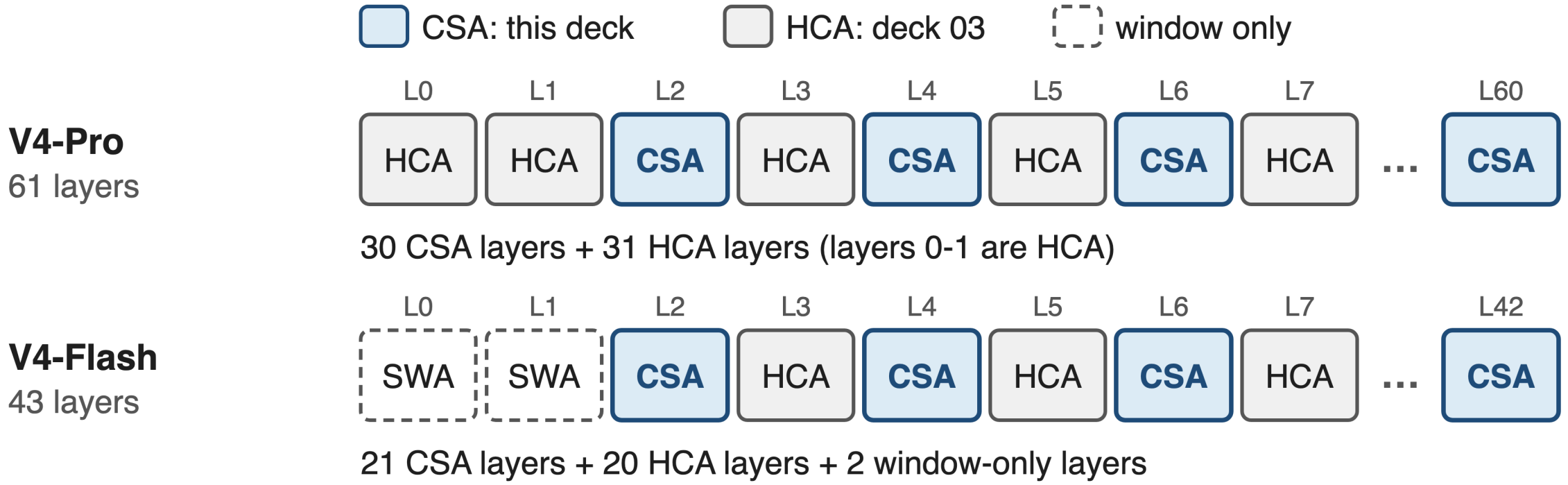
Select among compressed 4-token blocks, not tokens

DeepSeek-AI · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Attention & Long Context · arxiv.org/abs/2606.19348

V4 serves 1M tokens with two attention types



1M-token context (V3.2 stopped at 128K): CSA is the sparse half, HCA (deck 03) the dense half

At 1M, choosing costs more than attending

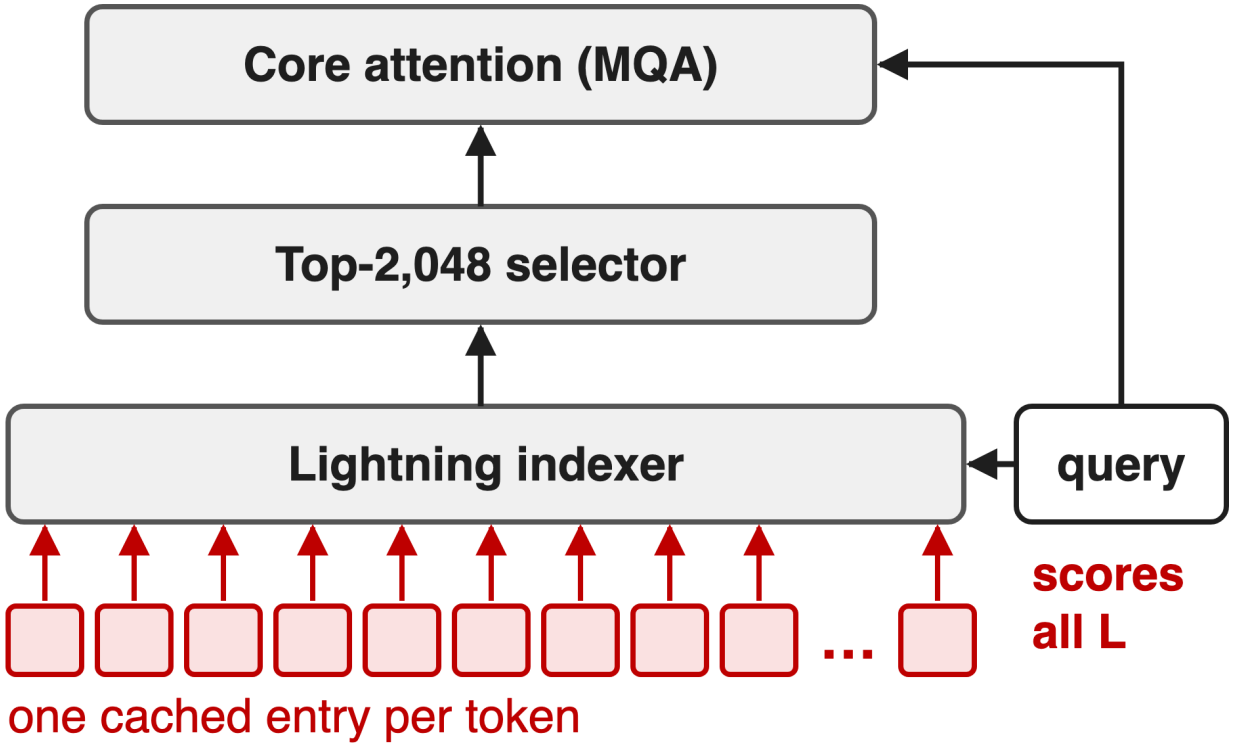
30×

more multiply-adds in DSA's indexer than in the attention it feeds

Analytic, per query per layer at 2^{20} tokens: indexer 64 × 128 × 1,048,576 = 8.6 G; attention over top-2,048 = 0.29 G (V3.2 §2.1, §2.3)

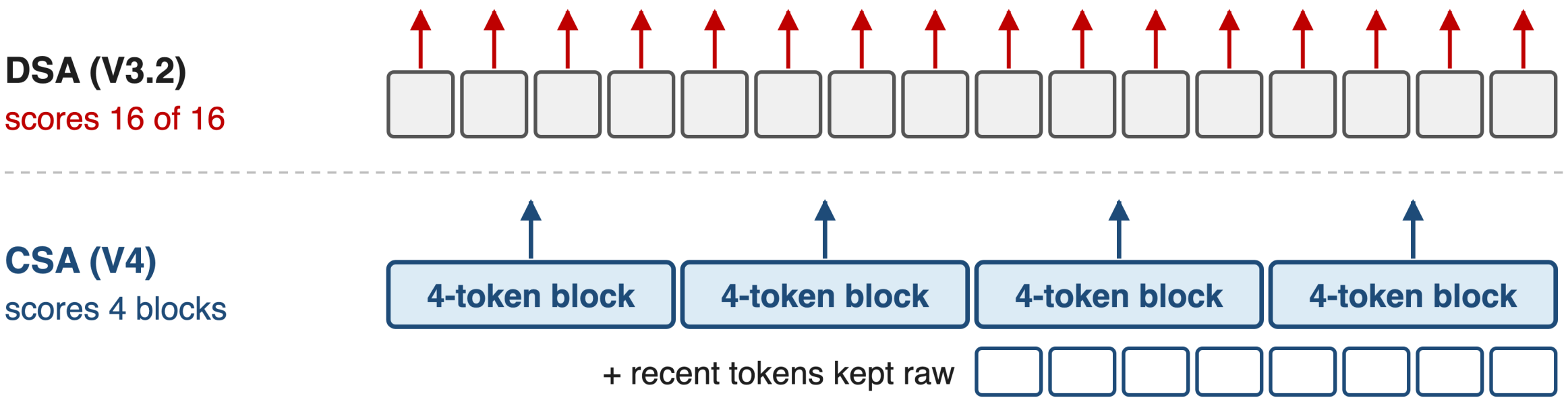
DSA selects tokens, so its cost tracks context

- **Dense MLA** (V3.1): reads all L entries
- **DSA** (V3.2): reads 2,048, still scores all L
- Cache: one 576-d entry per token, per layer



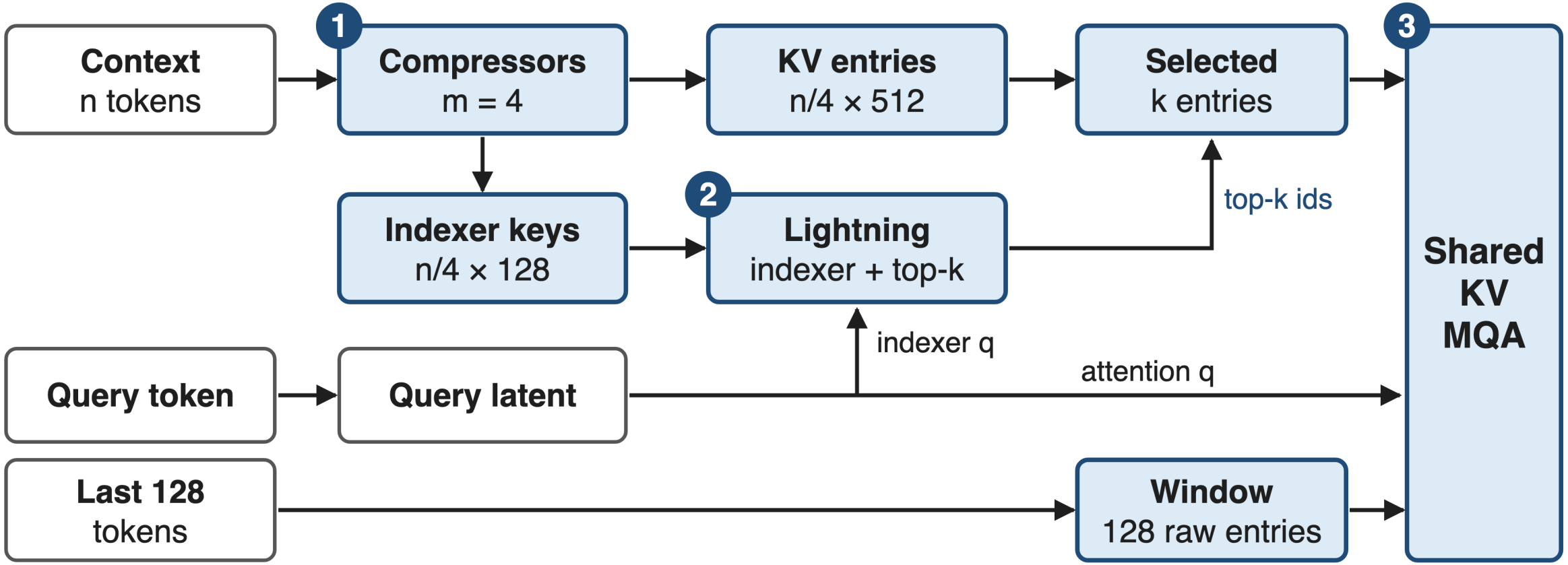
Villain: token granularity. Scan and cache both grow one-for-one with L

Key idea: select among blocks, not tokens



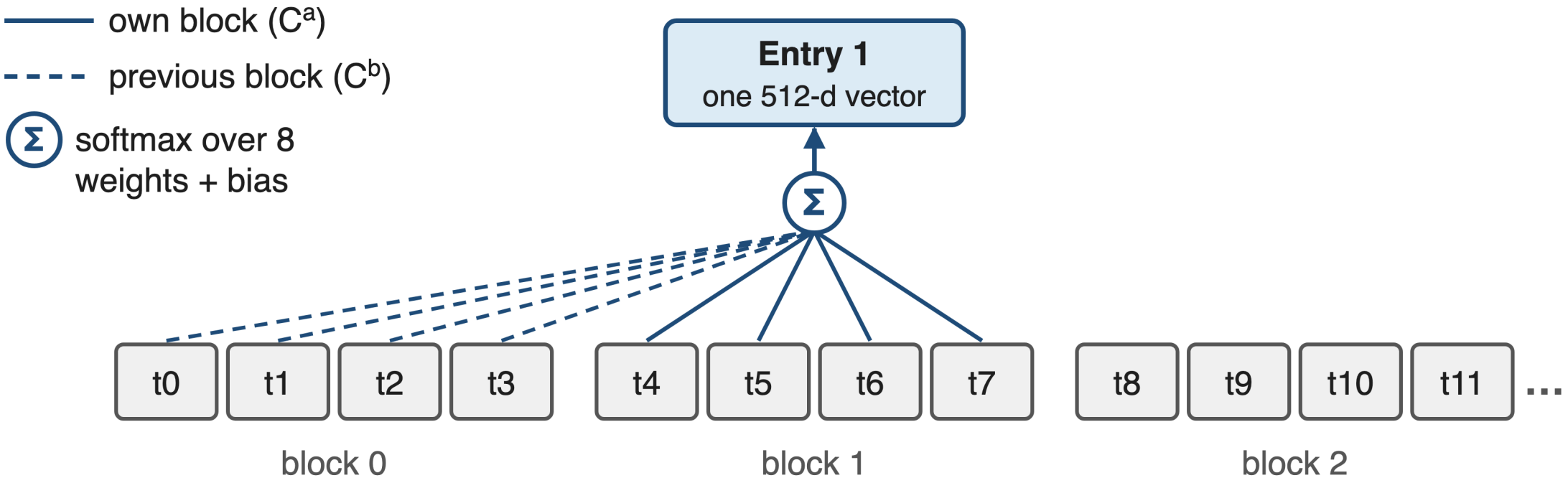
Score learned 4-token summaries: index scan and KV cache shrink 4×; a raw window keeps local detail

CSA compresses, indexes, then attends



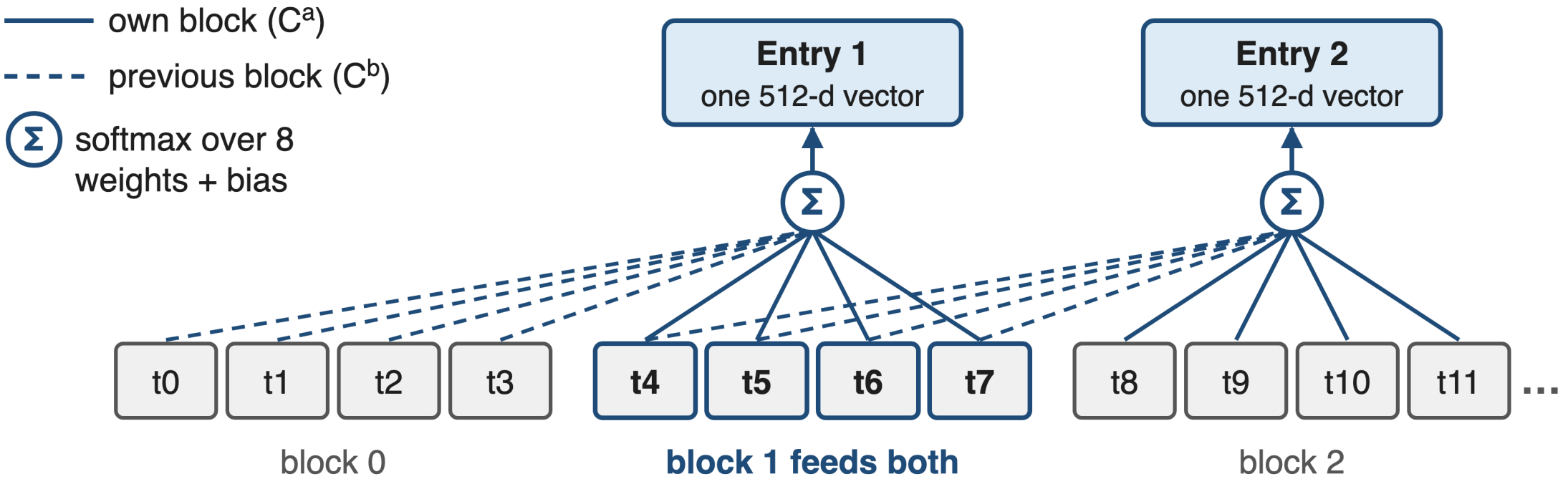
❶ compressors: n tokens → n/4 entries · ❷ indexer keeps the top-k blocks · ❸ MQA over k blocks + 128 raw tokens

❶ Learned pooling folds 4 tokens into one



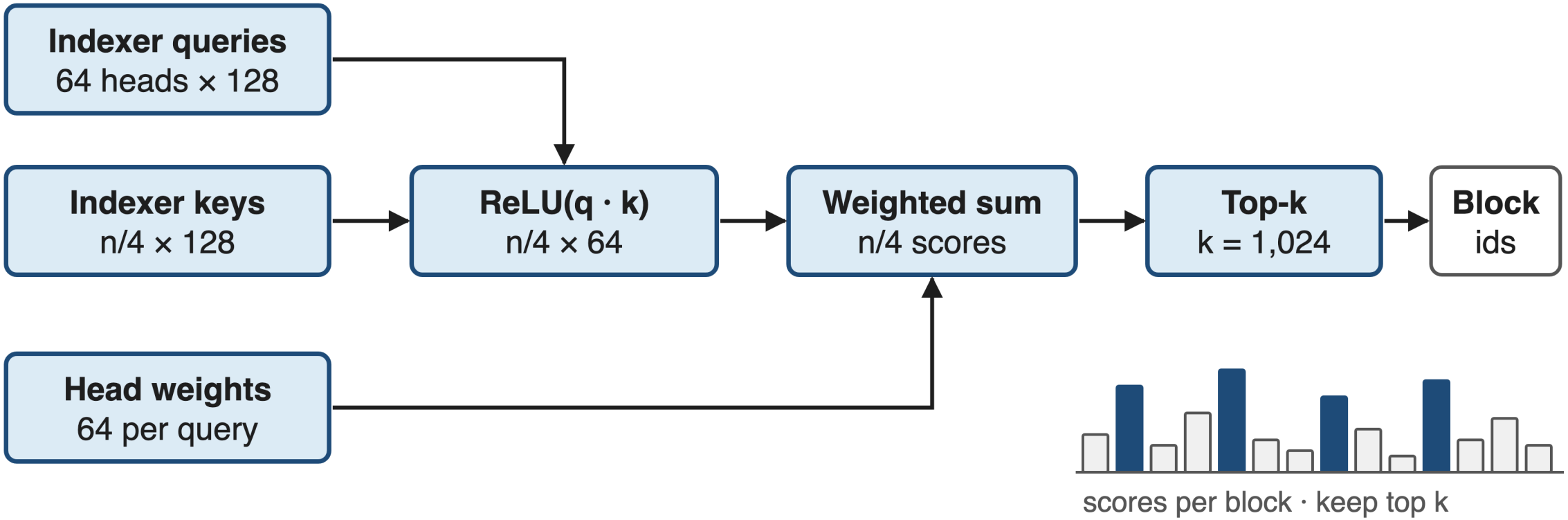
n tokens → n/4 entries of 512 dims; per-channel weights let each summary pick what to keep

❶ Learned pooling folds 4 tokens into one



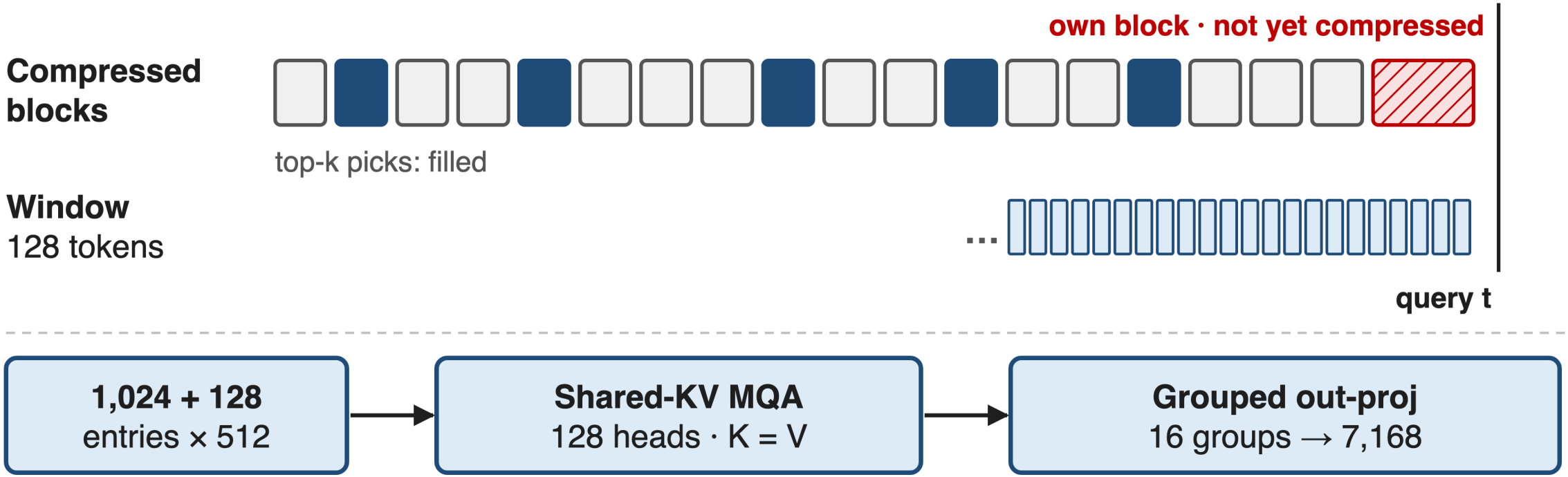
n tokens → n/4 entries of 512 dims; per-channel weights let each summary pick what to keep

❷ A light indexer keeps the top 1,024 blocks



64 heads × 128 dims = 8,192 multiply-adds per block; QK runs in FP4, scores in BF16 after QAT (deck 45)

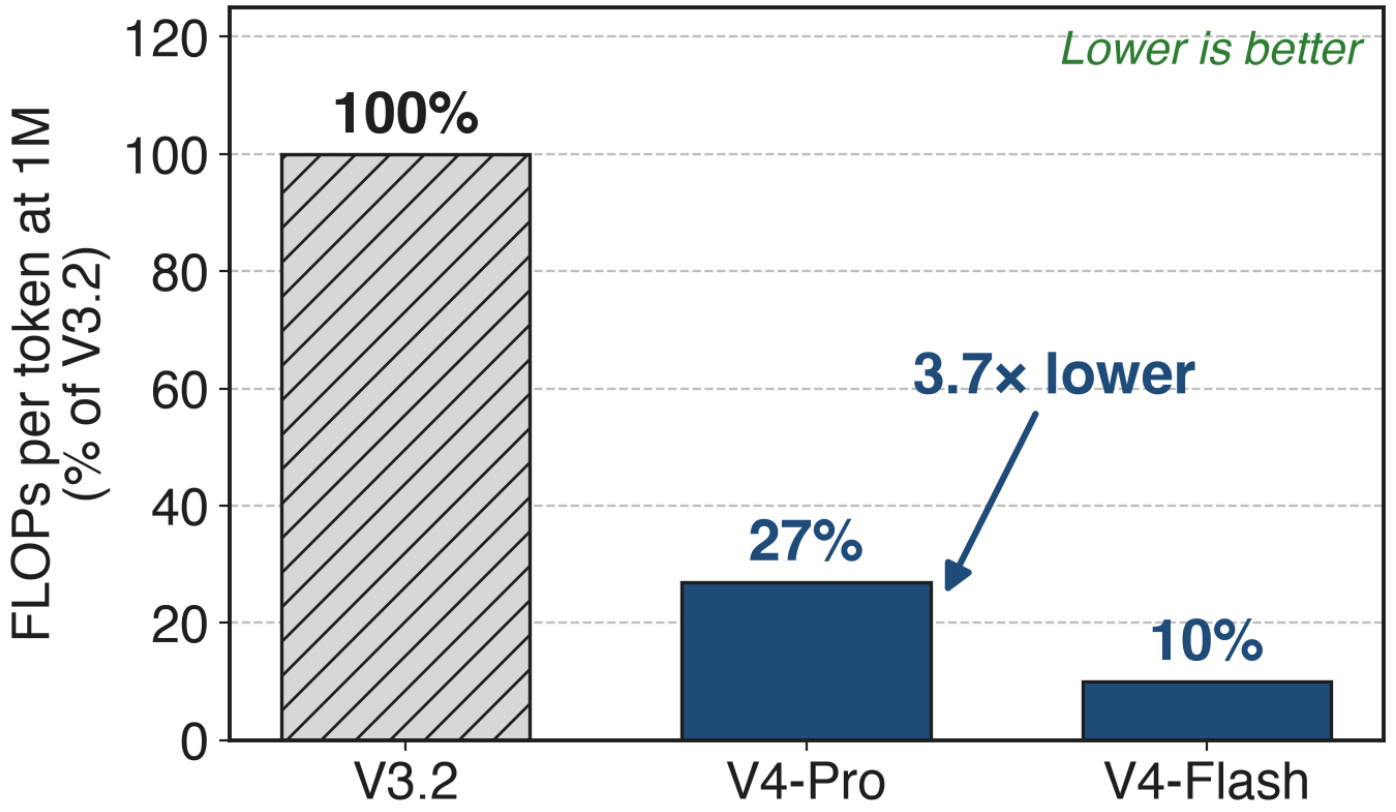
❸ A raw 128-token window restores locality



K = V = one 512-d entry: 1,152 entries feed 128 heads; outputs get RoPE(–i), so position stays relative

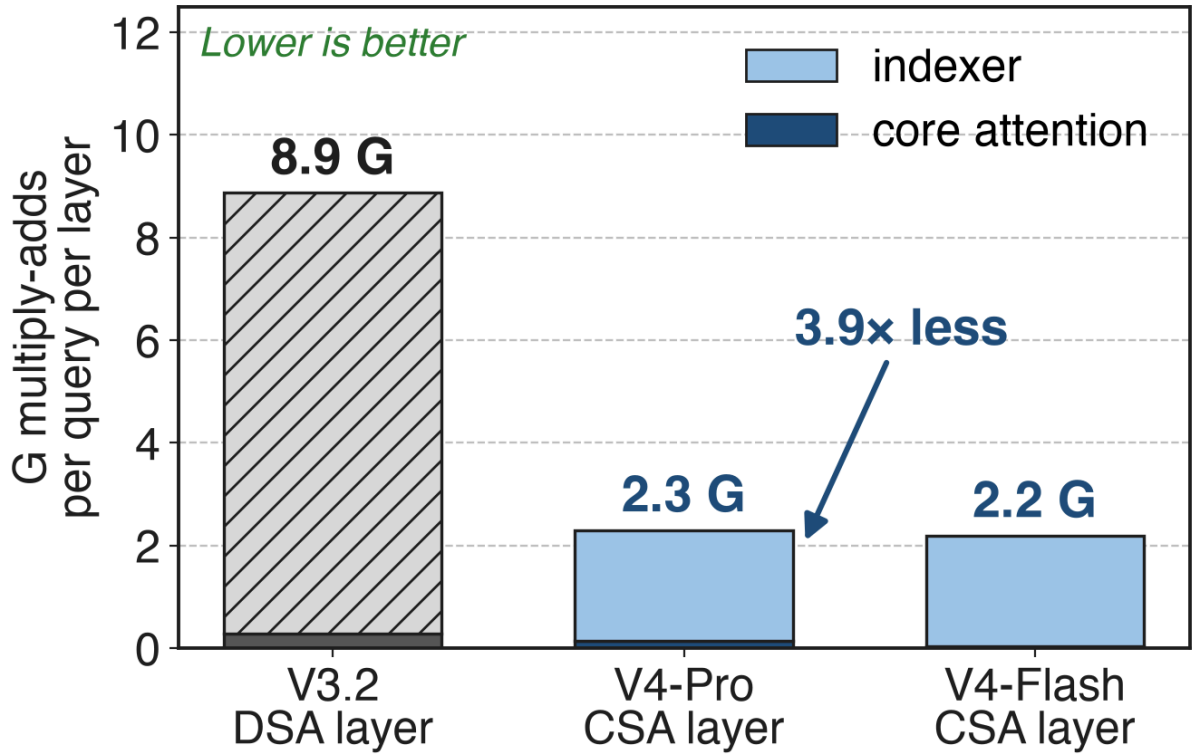
V4 needs 27% of V3.2's per-token FLOPs at 1M

- Whole V4 stack: CSA + HCA + low-precision storage
- Pro activates more: 49B vs 37B
- Flash: 10% (9.8× lower)



Per layer, CSA does 3.9× less attention work

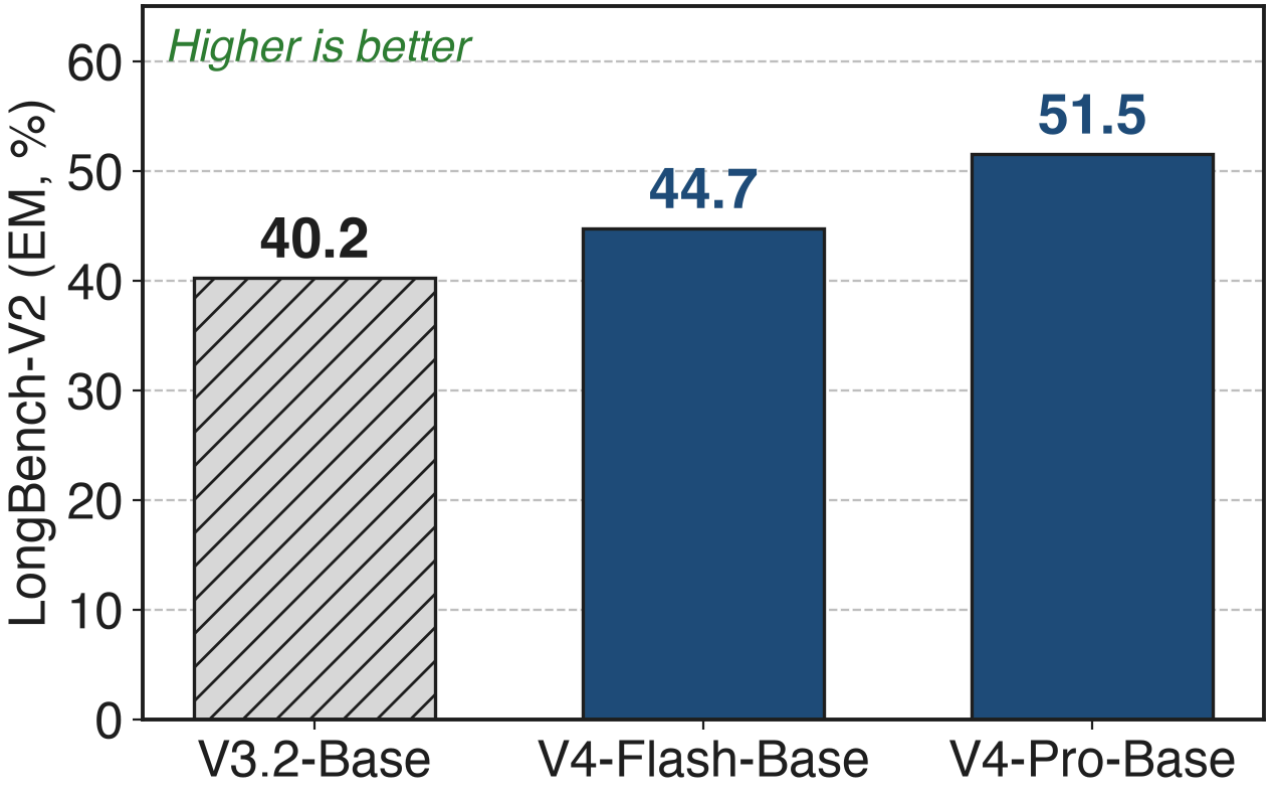
- Analytic: one query, 2^{20} tokens
- Indexer scan: 1,048,576 → 262,144
- Attended: 2,048 → 1,152 entries



The indexer still dominates, 14× core attention, which is why its QK path runs in FP4 (deck 45)

Long-context quality rises instead of falling

- MRCR at 1M: V4-Pro-Max **83.5** vs Gemini-3.1-Pro 76.3
- Claude Opus 4.6 still leads: 92.9
- Whole-model results: data and size changed too



Cost: a quadratic indexer and harder serving

- Indexer is still $O(L^2/4)$ in prefill and dominates each layer
- Training starts dense (Flash: first 1T tokens), then indexer warm-up, sparse from 64K
- Serving and CP: state cache for uncompressed tails, 128-token blocks, boundary exchange

No CSA-only ablation is published; DeepSeek calls the design "relatively complex"

Summary

- **Pain**: at 1M, DSA's indexer does 30× the attention's work (analytic)
- **Insight**: select among learned 4-token blocks, plus 128 raw tokens
- ❶ overlapping pooling ❷ FP4 indexer, top-1,024 ❸ shared-KV MQA
- V4 at 1M: **27%** / **10%** of V3.2's per-token FLOPs; LongBench-V2 **40.2** → **51.5**

https://arxiv.org/abs/2606.19348